

WARNING: THIS REVIEW IS NOT MEANT TO BE REFLECTIVE OF THE ACTUAL EXAM. THIS IS SIMPLY A SET OF SOME EXTRA PROBLEMS ON THE EXAM MATERIAL. IT IS NOT EXHAUSTIVE OF ALL TYPES OF MATERIAL OR QUESTIONS. YOU SHOULD USE THIS AS ONE STUDY TOOL, NOT YOUR ONLY STUDY TOOL!

During the test, no Excel formulas are allowed in your responses!

1. **MCQ.** What is the primary purpose of a random variable in probability theory?
 - a) To assign probabilities to each outcome in the sample space.
 - b) To map each outcome in the sample space to a real number.
 - c) To ensure that all outcomes in the sample space are equally likely.
 - d) To calculate the expected value of a probability distribution.

2. **MCQ.** What are the possible values of a Bernoulli random variable Z ?
 - a) Any real number between 0 and 1.
 - b) Only positive integers.
 - c) Only 0 and 1.
 - d) All positive and negative integers.

3. **True or False:** The pmf $p(x)$ is used to describe the probability distribution of a continuous random variable.

4. **Let X be the number of successful free throws a basketball player makes in a game. The pmf table of X is as follows:**

$X = x$	0	1	2	3	4	ow
$P(X = x)$	0.1	0.3	0.4	0.15	0.05	0

- a) What is the domain of X ?
 - b) Is X a discrete or a continuous variable?
 - c) Calculate the mean and the standard deviation of X and interpret.
 - d) Find $E(2X + 4)$ and $V(3X + 15)$.

5. **True or False:** In the binomial distribution probability mass function (pmf), the coefficient "n choose x" (denoted by $\binom{n}{x}$) represents the number of ways to choose x successes out of n trials.

6. A quality control engineer is inspecting a production line where light bulbs are manufactured. Historically, 5% of the bulbs produced on this line are defective. The engineer plans to randomly sample 20 light bulbs for inspection. Let X denote the number of defective sampled bulbs.
- Does X follow a binomial distribution? If so, give its parameters.
 - Calculate the probability that exactly 2 of the 20 sampled bulbs are defective.
 - Determine the probability that at most 3 of the sampled bulbs are defective.
7. In response to the availability of more resources and after a recent machinery maintenance, the quality control team decides to increase the sample size from 20 to 5000 for inspecting light bulbs produced on a manufacturing line. Assume now that only 0.01% of the bulbs are defective.
- Is it appropriate to use a Poisson distribution here to approximate the binomial distribution? If so, give its parameter.
 - Calculate the new probability that exactly 2 out of the 5000 sampled bulbs are defective.
 - Determine the probability that at most 3 of the sampled bulbs are defective.
8. An aerospace engineer is responsible for estimating the probability (p) of a critical component on a spacecraft failing during launch. In a series of 50 past launches, the component failed in 6 of them. **Estimate the probability (p) of the component failing during launch and determine the standard error for this estimate.**
9. **MCQ.** In a Poisson process, which of the following is a key assumption?
- Events occur at fixed, regular intervals.
 - Events are independent and identically distributed.
 - Events are always equally likely.
 - Events follow a specific pattern or structure.
10. **MCQ.** In a Poisson process, the rate parameter (λ) represents:
- The total number of events that will occur in a given time interval.
 - The time it takes for one event to occur.
 - The probability of an event happening.
 - The average rate at which events occur in a given time interval.

11. In a chemical manufacturing plant, the number of chemical reactions occurring per minute follows a Poisson distribution with an average rate of 5 reactions per minute.
- Calculate the probability that exactly 3 reactions occur in a 1-minute time frame.
 - What is the expected number of chemical reactions during a 5-minute time frame?
12. In a mechanical workshop, machine breakdowns occur according to a Poisson process with an average rate of 2 breakdowns per day. **Calculate the probability that there are no machine breakdowns in a particular 8-hour shift. Find the standard deviation and interpret it.**
13. **True or False:** The probability density function $f(x)$ describes the probability of the random variable X taking the specific value x .
14. **True or False:** The probability density function $f(x)$ is always non-negative, and its integral over its entire range equals 1 for a continuous random variable.
15. **True or False:** A continuous random variable has an infinite number of possible values, and the probability of any one particular value is zero.
16. **True or False:** The median of a random variable X with the probability density function described is the value of ' x ' for which the cumulative distribution function (cdf) reaches 0.5.
17. In the field of **materials engineering**, a specific parameter, denoted as **X** , represents the **tensile stress experienced by a material during a testing process**. The behavior of this parameter follows a probability density function (pdf) characterized by the following expression:

$$f(x) = \begin{cases} cx^2 & |x| \leq 1 \\ 0 & otherwise \end{cases}$$

- To ensure accurate stress analysis, engineers need to **determine the constant 'c'** that characterizes the pdf for material stress. This constant is crucial in understanding the behavior of the material under test conditions.
- Engineers are interested in **calculating the expected value (mean)** of the material stress, represented as $E(X)$. This allows them to predict the typical or average stress the material experiences during testing.

- c) In materials engineering, understanding the variability in stress levels is essential. Engineers need to **determine the variance, $V(X)$** , to quantify the spread or dispersion of stress values.
- d) When designing materials for specific applications, it's crucial to assess extreme scenarios. Engineers aim to find the probability that material stress (X) exceeds or equals a critical value of $\frac{3}{4}$, denoted as **$P(X \geq \frac{3}{4})$** . This information guides material selection for applications with stringent stress requirements.
- e) Determine the **median (Q_2)** of the material stress distribution represented by the pdf.
- f) Calculate the **first quartile (Q_1)** of the material stress distribution.

18. True/False: The rate parameter λ of an exponential distribution represents the average number of events occurring in a unit of time.

19. MCQ. What does it mean for a probability distribution to have the memoryless property?

- a) It remembers past events.
- b) The probability of an event occurring depends on past events.
- c) It is unaffected by past events.
- d) It is always deterministic.

20. MCQ. In the context of the memoryless property, which of the following distributions exhibits this property?

- a) Normal distribution
- b) Exponential distribution
- c) Poisson distribution
- d) Uniform distribution

21. MCQ. The exponential distribution is often used to model the time between events in a Poisson process. Which statement is true regarding the memoryless property in this context?

- a) The longer you wait, the more likely the event is to occur.
- b) The longer you wait, the less likely the event is to occur.
- c) The probability of the event occurring is the same, regardless of how much time has passed.
- d) The probability of the event occurring increases with time.

- 22. MCQ.** If an exponential distribution exhibits the memoryless property, what can you say about the probability of an event occurring in the next unit of time, given that it hasn't occurred in the previous unit of time?
- a) The probability increases.
 - b) The probability decreases.
 - c) The probability remains the same.
 - d) It's impossible to determine.
- 23. (The Waiting Time Paradox)** In Fishtown, buses do not operate on a fixed schedule. Instead, they arrive according to a Poisson process at a rate of one per 10 minutes, (i.e., $\lambda = 0.1$ arrivals per minute. How long do you have to wait before the next bus arrives?
- 24. MCQ.** Which of the following statements about the normal distribution is true?
- a) It is a discrete probability distribution.
 - b) It is always skewed to the left.
 - c) It is defined by its mean and standard deviation.
 - d) It has a fixed range of values.
- 25. MCQ.** In a standard normal distribution (mean = 0, standard deviation = 1), what percentage of data falls within one standard deviation of the mean?
- a) 25%
 - b) 50%
 - c) 68%
 - d) 95%
- 26. MCQ.** What is the shape of the probability density function (pdf) for a normal distribution?
- a) Uniform
 - b) Exponential
 - c) Bell-shaped
 - d) Skewed
- 27. MCQ.** In a normal distribution, approximately what percentage of data falls within two standard deviations of the mean?
- a) 34%
 - b) 50%
 - c) 68%
 - d) 95%

- 28. MCQ.** The Empirical Rule, also referred to as the 68-95-99.7 rule, applies to which type of distribution?
- a) Exponential distribution
 - b) Uniform distribution
 - c) Normal distribution
 - d) Poisson distribution
- 29. MCQ.** What is the main characteristic of a z-score in the context of a normal distribution?
- a) It represents the number of standard deviations a data point is from the mean.
 - b) It is the probability associated with a specific data point.
 - c) It quantifies the degree of skewness in the distribution.
 - d) It is the median value of the distribution.
- 30.** The scores on a standardized college admission test follow a normal distribution with a mean (μ) of 500 and a standard deviation (σ) of 100. You are asked to find the following probabilities:
- a) What is the probability that a randomly selected student scores above 600 on the test?
 - b) What is the probability that a randomly selected student scores below 450 on the test?
 - c) Find the z-score for a student who scores 550 on the test.
 - d) If the top 10% of students are to be selected for a special program, what is the minimum score required for eligibility?
- 31. True or False:** The upper and lower percentiles of a normal distribution are symmetrical with respect to its mean. For example, if you have the upper 10th percentile on the right side of the mean, and the lower 10th percentile on the left side, will they have the same percentage of data?
- 32. True or False:** A z-score can be used to compare data points from two different datasets, even if the datasets have different units of measurement.
- 33. True or False:** A z-score can be calculated for any type of data distribution, regardless of its shape.

- 34.** In a manufacturing process, the diameter of a certain component is normally distributed with a mean (μ) of 40 millimeters and a standard deviation (σ) of 2 millimeters.
- What is the probability that a randomly selected component has a diameter between 38 and 42 millimeters?
 - What is the probability that a randomly selected component has a diameter less than 36 millimeters?
 - If the specifications require that 95% of the components have diameters within a certain range, what is the upper and lower limit for the diameter of these components?
- 35.** Suppose a standardized test has a mean (μ) score of 150 and a standard deviation (σ) of 30. A student scores 180 on the test. **Calculate and interpret the z-score of the standardized test score.**
- 36. MCQ.** What is the main purpose of a normal probability plot?
- To create a bar graph of data distribution.
 - To display data in a pie chart.
 - To assess the normality of a dataset.
 - To calculate the mean and standard deviation of data.
- 37. True or False:** A normal probability plot is the only graphical method that helps assess whether a dataset follows a normal distribution.
- 38. MCQ.** In a normal probability plot, what is expected to happen if the data closely follows a straight line?
- It indicates data is not normally distributed.
 - It suggests data follows a uniform distribution.
 - It suggests data follows a normal distribution.
 - It has no specific meaning.
- 39. MCQ.** What does a systematic deviation from a straight line, such as an S-curve pattern in a normal probability plot, typically indicate?
- Data is normally distributed.
 - Data follows an exponential distribution.
 - Data does not follow a normal distribution.
 - Data is symmetrical.
- 40. MCQ.** What do you compare in a normal probability plot to assess the normality of a dataset?
- Observed data points to an imaginary line.
 - Observed data points to an expected straight line.
 - Observed data points to a pie chart.
 - Observed data points to a bar graph.

- 41. True or False:** When the sample size is small, a histogram may not provide an accurate assessment of the normality of the data.
- 42. MCQ.** What is the key difference between the population distribution and the sampling distribution?
- a) Population distribution characterizes the entire group, while sampling distribution characterizes some aspect of sampling variability.
 - b) Population distribution is always normal, while sampling distribution can be any shape.
 - c) Population distribution is always smaller than the sampling distribution.
 - d) Population distribution is created from samples, while sampling distribution is the complete dataset.
- 43. MCQ.** Sampling variability refers to:
- a) The process of selecting a random sample from the population.
 - b) Variability among random samples from the same population.
 - c) The range of values in the population distribution.
 - d) The sampling bias introduced during data collection.
- 44. MCQ.** In sampling theory, what is the purpose of random sampling?
- a) To ensure that the sampling distribution is identical to the population distribution.
 - b) To introduce bias into the sample.
 - c) To reduce the chances of obtaining a representative sample.
 - d) To eliminate selection bias and make the sample more representative of the population.
- 45. MCQ.** If you increase the sample size in a study, how does it typically affect sampling variability?
- a) It reduces sampling variability.
 - b) It increases sampling variability.
 - c) It has no impact on sampling variability.
 - d) It makes the population distribution more skewed.
- 46. MCQ.** What does the Central Limit Theorem state?
- a) It explains how to calculate the mean of a dataset.
 - b) It asserts that the population mean is equal to the sample mean.
 - c) It describes the distribution of sample means as approximately normal, regardless of the population distribution.
 - d) It defines the standard deviation of a population.

- 47. MCQ.** According to the Central Limit Theorem, for a sufficiently large sample size, the sampling distribution of the sample mean becomes approximately:
- a) Uniform
 - b) Exponential
 - c) Normal
 - d) Poisson
- 48. MCQ.** What is the main benefit of the Central Limit Theorem in statistics?
- a) It allows us to compute the standard deviation of a sample.
 - b) It ensures that the population distribution is always normal.
 - c) It provides a method for estimating population parameters.
 - d) It allows us to make inferences about the population mean regardless of the population distribution.
- 49. MCQ.** The Central Limit Theorem assumes that the samples are drawn from a population with a finite mean and finite:
- a) Variance
 - b) Median
 - c) Mode
 - d) Range
- 50. MCQ.** In practical terms, when is it often safe to apply the Central Limit Theorem?
- a) For any sample size.
 - b) Only when the population distribution is normal.
 - c) When the sample size is sufficiently large (e.g., typically $n \geq 30$).
 - d) Only when the sample size is very small (e.g., $n < 5$).
- 51.** In a chemical manufacturing process, the concentration of a certain compound in the final product is known to be normally distributed with a mean concentration (μ) of 25 grams per liter and a standard deviation (σ) of 4 grams per liter. To ensure quality control, samples of 25 liters of the final product are taken.
- a) Calculate the mean concentration and standard error of the sample mean.
 - b) What is the distribution of the sample mean?
 - c) Does the Central Limit Theorem apply here?
 - d) Calculate the probability that the sample mean concentration is less than 24 grams per liter.
 - e) Calculate the probability that the sample mean concentration is greater than 26 grams per liter.
- 52.** In a packaging facility, the weights of bags of a certain product follow a population distribution with a mean weight (μ) of 200 grams and a standard deviation (σ) of 10 grams. To ensure quality control, random samples of 64 bags are taken.
- a) Calculate the mean and standard error of the sampling distribution of the sample means.

- b) Determine the probability that the sample mean weight is less than 198 grams.
- c) Calculate the probability that the sample mean weight is greater than 202 grams.
- d) What is the z-score for a sample mean weight of 198 grams?
- e) What is the z-score for a sample mean weight of 202 grams?
- f) Interpret the z-scores calculated in questions d) and e) in the context of this problem.

53. MCQ. In a t-distribution, as the degrees of freedom (df) increase:

- a) The distribution becomes more symmetrical and approaches a normal distribution.
- b) The distribution becomes more skewed.
- c) The distribution becomes flatter and wider.
- d) The distribution remains unchanged.

54. MCQ. What is the parameter for defining a specific t-distribution?

- a) Sample size (n)
- b) Population mean (μ)
- c) Degrees of freedom (df)
- d) Standard deviation (σ)

55. MCQ. In a t-distribution, if the degrees of freedom are 10, what is the shape of the distribution compared to a normal distribution?

- a) T-distribution is always more symmetrical than a normal distribution.
- b) T-distribution is always more skewed than a normal distribution.
- c) T-distribution is similar to a normal distribution but narrower.
- d) T-distribution is similar to a normal distribution but has heavier tails.

56. The lifetimes of a particular brand of light bulbs are normally distributed with a mean lifetime (μ) of 1200 hours and an unknown population standard deviation. A random sample of 15 light bulbs is selected, and their lifetimes are recorded. Calculate the following:

- a) The degrees of freedom for the t-distribution in this scenario.
- b) The 95th percentile of the t-distribution with the calculated degrees of freedom.
- c) The probability that the sample mean lifetime is less than 1150 hours. The estimated sample standard deviation is $s = 100.5$ hours.

57. MCQ. What does the shape of the chi-squared distribution depend on primarily?

- a) Degrees of freedom
- b) Sample size
- c) Mean and standard deviation
- d) Population size

- 58.** A quality control manager is conducting tests on the thickness of a certain type of glass sheets produced in a factory. The thickness measurements follow a chi-squared distribution with 10 degrees of freedom. The manager needs to establish quality control limits.
- a) Calculate the 90th percentile of the chi-squared distribution with 10 degrees of freedom.
 - b) Determine the probability that a randomly selected glass sheet will have a thickness measurement exceeding the quality control limit defined by the 90th percentile.
- 59.** A statistics instructor is grading final exams. The instructor is interested in the distribution of exam scores, which follows an F-distribution with 3 and 15 degrees of freedom for the numerator and denominator, respectively. The instructor wants to set a threshold for the top 10% of scores.
- a) Calculate the 90th percentile of the F-distribution with 3 and 15 degrees of freedom.
 - b) Determine the threshold for the top 10% of scores based on the calculated percentile.
- 60. True or False:** Degrees of freedom are a measure of the number of data points in a dataset.